# Full-Scale Change Detection Network for Remote Sensing Images Based on Deep Feature Fusion

Shenbo Liu, Dongxue Zhao, Yuheng Zhou, Ying Tan, Huang He, Zhao Zhang, and Lijun Tang

***Abstract*— In the processing of high-resolution remote sensing images, multiscale feature fusion techniques are commonly employed to construct change detection models, aiming to capture the details and characteristics of target objects at different scales. However, current change detection methods often neglect the subtle features of small-scale targets and edge information during feature fusion. To address this issue, this article proposes a deep feature fusion network (DFFNet) for full-scale change detection in remote sensing images. DFFNet enhances the accuracy of boundary information in change regions and enables the extraction of full-scale change features. By integrating phased difference extraction techniques with an enhanced attention mechanism, a dual temporal difference enhancement module (DT-DEM) is designed to comprehensively extract change detection information across scales. In addition, combining an inverted pyramid technique with an attention mechanism, a screening function inverted pyramid network (S-FIPN) is proposed, which efficiently extracts and fuses full-scale features while significantly improving the extraction of subtle changes and weak edge features. Comparison experiments with ten state-of-the-art (SOTA) algorithms on four publicly available datasets, namely, LEVIR-CD, WHU-CD, NJDS, and MSRS-CD, show that DFFNet achieves the best $F1$ values on all datasets, which are 92.50%, 90.55%, 81.04%, and 77.44%, respectively. These results underscore the superiority and effectiveness of DFFNet in full-scale feature extraction and edge information retrieval.**

***Index Terms*— Change detection, deep feature fusion, dual-temporal difference enhancement, high-resolution remote sensing images.**

## I. Introduction

CHANGE detection refers to the identification and extraction of changes in content from two or more images acquired at different times over the same geographical area [1]. Currently, change detection technology based on high-resolution remote sensing images has been widely applied in fields such as urban planning [2], [3], environmental monitoring [4], [5], and disaster assessment [6], [7].

Traditional change detection methods are primarily categorized into pixel-based and object-based approaches. Pixel-based change detection [8], [9] involves comparing the grayscale or color values of each pixel across two temporally aligned images after precise geometric registration. This approach determines whether a pixel has changed and identifies the regions of change. With its simplicity and ease of understanding, pixel-based methods perform well when processing medium- and low-resolution remote sensing images. However, for high-resolution remote sensing images, these methods are prone to generating noise-like pseudo-change information, which significantly impacts the efficiency and accuracy of change detection. On the other hand, object-based change detection [10], [11] first identifies objects of interest and extracts their features, then evaluates changes by comparing these features. Nevertheless, the uncertainty in object delineation during image segmentation limits the performance of object-based change detection.

In recent years, change detection methods based on deep learning have become a research hotspot due to their powerful modeling and learning capabilities. Deep learning models can extract complex, nonlinear, multilevel features directly from raw data, effectively modeling the relationship between images and actual geographic changes, thereby achieving more accurate change detection. These methods are characterized by strong robustness and high efficiency and have been widely applied in the field of high-resolution remote sensing image change detection. Currently, Siamese networks are the mainstream architecture for high-resolution remote sensing image change detection. However, most algorithms rely solely on the output of final feature maps for prediction, which makes it difficult to fully capture multiscale change features. To address this limitation, researchers have introduced feature fusion techniques into change detection tasks to achieve multiscale feature integration. For instance, Zheng et al. [12] designed CLNet by incorporating multiscale fusion techniques into the encoder to facilitate multilevel feature integration. Similarly, Wang et al. [13] proposed ADS-Net, which combines attention mechanisms to achieve multiscale feature fusion. However, existing methods often overlook the extraction of small target features and edge information during the feature fusion process.

To address this issue, this article proposes a high-resolution remote sensing image full-scale change detection network based on a deep feature fusion network (DFFNet). The approach begins with a shared feature extraction network to preliminarily analyze and process the input bi-temporal images, extracting the fundamental feature information. Using these fundamental features, the network captures change features across different scales and enhances full-scale change information. Before feature fusion, a dual temporal difference enhancement module (DT-DEM) is introduced to refine the

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representation of change features. In addition, a screening function inverted pyramid network (S-FIPN) is designed to achieve multiscale change feature fusion. Finally, a change detection head is incorporated to generate the change mask. The main contributions of this article are as follows.

1) A DT-DEM is designed, which integrates attention mechanisms with phased difference extraction techniques to enhance the bi-temporal differences in remote sensing images. This module effectively extracts and amplifies full-scale change features.
2) A S-FIPN is proposed to extract and fuse full-scale change features while enhancing subtle edge changes, thereby improving detection performance.
3) Combining DT-DEM and S-FIPN, a full-size change detection network (DFFNet) based on deep feature fusion for high-resolution remote sensing images is designed, which is capable of extracting more accurate full-size change area and boundary information. Experiments on four datasets, LEVIR-CD, WHU-CD, NJDS, and MSRS-CD, show that DFFNet has significant advantages in full-size change feature extraction and edge information extraction compared with eight state-of-the-art (SOTA) algorithms.

The remainder of this article is organized as follows. Section II briefly reviews related work. In Section III, our proposed network architecture and detailed design are introduced. Then, experimental analysis is presented in Section IV. The generalization and limitations of the approach are discussed in Section V. Finally, the full article is summarized in Section VI.

## II. Related Work

### A. Deep Learning-Based Change Detection

Convolutional neural networks (CNNs) [14] and Transformer [15] architectures demonstrate powerful feature extraction capabilities and play a significant role in change detection tasks. CNNs, with their unique convolutional and pooling layers, automatically extract local features from images and progressively build a global understanding through stacked layers. This enables them to extract features from bi-temporal images and establish change detection models using mechanisms like feature differencing and dual-branch networks. Daudt et al. [16] utilized fully convolutional networks (FCNs) to extract change features, achieving early success in deep learning-based change detection. Lin et al. [17] proposed BCNN, a bilinear CNN-based method, which revealed relationships between bi-temporal feature maps by designing combined bilinear features.

With the deeper research on CNN-based change detection, researchers introduced an attention mechanism in the network to further improve the model's attention to fine-grained features and avoid the influence of external noise. Shi et al. [18] integrated a convolutional block attention module (CBAM) [19] into a metric learning module to design DSAMNet, enabling the acquisition of more discriminative features. Similarly, Chen and Shi [20] proposed STANet, which leveraged temporal and spatial attention mechanisms to capture more distinctive features while reducing the influence of pseudo-changes.

To enhance the global feature modeling capability of CNN, Chen et al. [21] proposed BIT by combining CNN and Transformer, which successfully achieved the modeling of contextual information in the temporal and spatial domains. Building on BIT, Jiang et al. [22] introduced VcT, which incorporated graph neural networks to model structured information and identify significant change features between two images. Zhang et al. [23] designed a pure Transformer-based change detection network, SwinSUNet, further enhancing the model's global feature extraction ability. Bandara and Patel [24] developed ChangeFormer by employing hierarchical Transformers and difference feature extraction modules, significantly improving change detection performance.

Despite these advancements, deep learning-based change detection methods face limitations in associating and fusing features across different scales. They also struggle with detecting small objects and weak edge features, particularly in applications involving high-resolution remote sensing images.

### B. Multiscale Feature Fusion

Multiscale feature fusion methods can be broadly categorized into three approaches: attention mechanism-based, CNN-based with different receptive fields, and pyramid-based.

Attention-based methods leverage attention mechanisms to enhance feature representation across different scales. Song et al. [25] proposed CSANet, which incorporates cross-temporal attention to improve feature representation at various stages. Cheng et al. [26] designed H-SALENet with a hierarchical attention mechanism, enhancing the representation of hierarchical and long-range dependency features. Peng et al. [27] introduced DDCNN by combining spatial and channel attention mechanisms, significantly improving the representation of fine-grained changes. However, these methods often focus on prominent features, potentially overlooking small target features or weak edge features, especially when processing high-resolution images.

Based on CNNs with different receptive fields. By utilizing different kernel sizes and jump connections, such methods are able to capture multiscale features within network layers. Gu et al. [28] proposed the dense difference fusion module to allow the network to detect multiple changing objects of different sizes through full-size hopping connection aggregation. Ding et al. [29] proposed an attention-guided cross-layer connection module to achieve multiscale feature fusion. While skip connections facilitate the integration of features from different levels, their ability to deeply interact with multiscale features remains limited, making it challenging to fully leverage the synergy between scales.

Pyramid-based methods capture rich spatial details by creating and processing multiscale input images. Jiang et al. [30] designed PGA-SiamNet based on a feature pyramid to achieve multiscale feature fusion. Chen et al. [31] constructed a feature pyramid with an asymmetric interactive attention module, further enhancing feature fusion performance. However, these methods often require multiple scale resizings of the input

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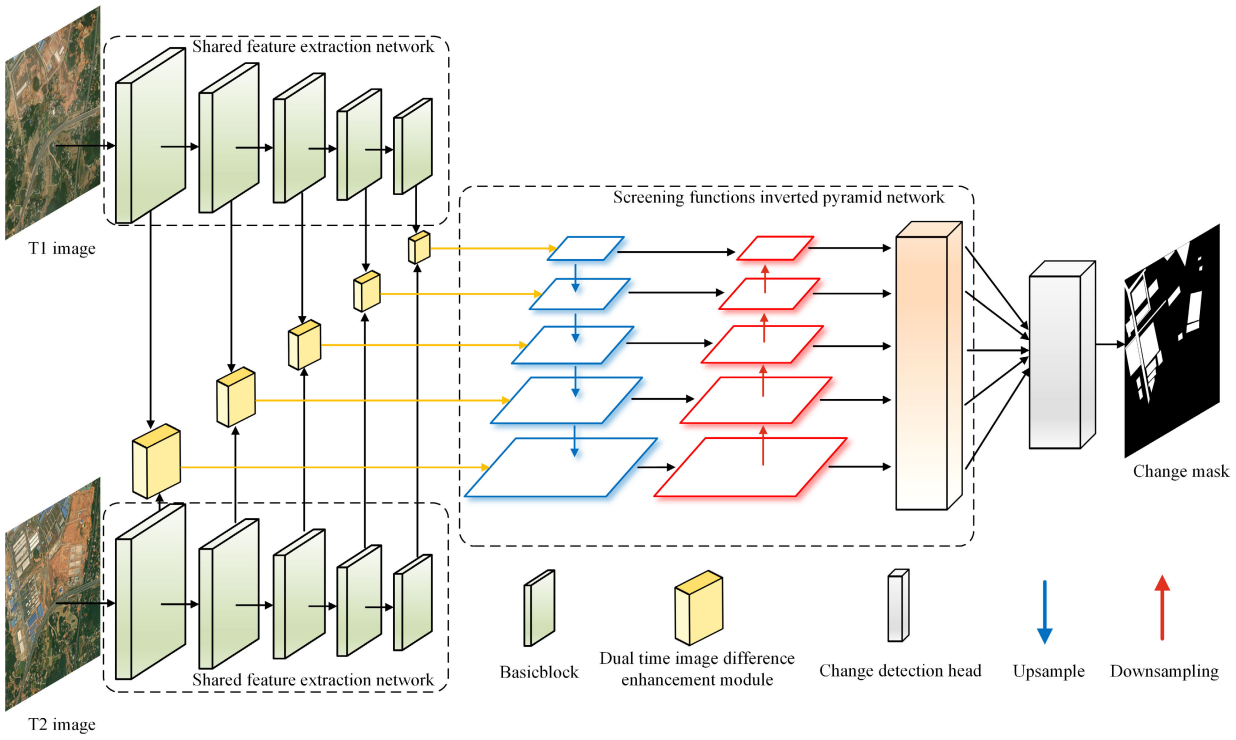



Fig. 1. DFFNet network structure diagram, DFFNet consists of five parts: input is a pair of images, output is a change mask, contains a shared feature extraction network, a dual time image difference enhancement module, a screening functions inverted pyramid network, and a change detection head.

images, leading to potential information loss or noise introduction, particularly in high-resolution image processing.

To address the limitations of existing multisize feature fusion methods for change detection, this article designs a DT-DEM module that enhances small size and edge feature extraction to realize multisize feature extraction before feature fusion. At the same time, the S-FIPN module is developed, which significantly enhances the extraction of small change and edge features and realizes the feature fusion and extraction of change features of different sizes.

## III. Methodology

### A. Overall Framework

The network structure of DFFNet is shown in Fig. 1. The preliminary analysis and processing of the input images of the two time-images are carried out through the shared feature extraction network to extract the basic feature information in the images, which provides the data basis for the subsequent processing steps. The extracted preliminary features are then passed to the dual time image difference enhancement module, which extracts change features across five different scales. These multiscale change features are subsequently input into the screening functions inverted pyramid network. This module conducts more in-depth change feature detection and refinement while achieving multiscale feature fusion, yielding more precise change information. Finally, a change mask is generated by the change detection header. This change mask can clearly and accurately mark all the changed areas in the original input image, thus realizing the effective identification and labeling of the changed areas in the image.

### B. Shared Feature Extraction Network

The change detection tasks typically rely on bi-temporal image data and commonly utilize feature extraction networks with weight-sharing characteristics to extract features from the two temporal images. Traditional shared feature extraction networks can be broadly categorized into two types: fully convolutional Siamese networks and Transformer-based Siamese networks. These networks often adopt well-established feature extraction models, such as ResNet [32], VGG [33], or Transformer [15], as their backbone. However, these mature networks are typically designed for input sizes of 224 × 224, whereas remote sensing image datasets generally have dimensions of 1024 × 1024. Directly upscaling features from 224 × 224 to 1024 × 1024 through interpolation or tiling can introduce stitching errors and lead to inconsistencies between local and global features. In addition, the high-resolution nature of 1024 × 1024 images provides richer global information, and its feature representation requirements are significantly higher than those of smaller 224 × 224 images. To address these challenges, DFFNet sets its input size to 1024 × 1024 to fully leverage the detailed and global features of high-resolution remote sensing images, ensuring accurate and consistent change detection.

Based on this, this article constructs the feature extraction network with the basic module shown in Fig. 2. Fig. 2(a) is the basic module with downsampling function, and Fig. 2(b) is the basic module without downsampling. Feature extraction is performed through the CBR module, which utilizes CBAM [19] to aggregate features to better highlight object boundaries and suppress background features. Fig. 2(a) is used to obtain five different scales of features, which are $(h/2, w/2)$, $(h/4, w/4)$,

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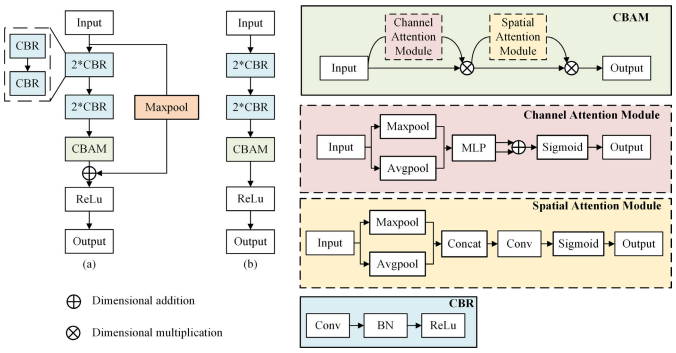



Fig. 2. Large-size bi-temporal difference enhancement module structure. The basic module of the feature extraction network, in which (a) is the basic module with downsampling function and (b) is the basic module without downsampling.

$(h/8, w/8)$, $(h/16, w/16)$, and $(h/32, w/32)$, where $h$ represents the height of the input image and $w$ represents the width of the input image. The two-time image difference enhancement module in Fig. 1 is utilized to extract the corresponding five scales of variation information from the feature information of these five scales.

### C. Dual Time Image Difference Enhancement Module

Existing change detection networks rely on direct difference operations between dual-time images to extract change features. Thanks to the strong semantic representation capabilities of high-level features, these networks can effectively represent high-level semantic information such as object categories and scene types in images. This approach performs well when processing high-level features. However, low-level features, which contain rich raw spatial information such as edges, textures, and color details, are difficult to comprehensively and accurately capture using direct difference operations. To simultaneously extract high-level change information and accurately capture changes in low-level features, this article designs an enhanced attention mechanism and proposes the DT-DEM to improve the accuracy and robustness of change detection.

The structure of the enhanced attention mechanism is shown in Fig. 3. This module is an improvement upon the CBAM. The channel attention mechanism of CBAM performs channel dimensionality reduction through a multilayer perceptron (MLP), which may have a negative impact on channel attention prediction. In addition, capturing dependencies between all channels can be somewhat redundant. To address this, the channel attention mechanism of CBAM is replaced with efficient channel attention (ECA) [34]. ECA uses 1-D convolution to achieve local cross-channel interaction without dimensionality reduction, which not only reduces the number of parameters but also significantly improves performance.

The spatial attention mechanism follows the design of CBAM, but the enhanced attention mechanism introduces an adaptive weight mechanism with learnable parameters, enabling the dynamic fusion of the outputs from both channel and spatial attention. This improvement avoids the suppression of detailed information that may occur during direct multiplication, especially when the contributions of channel and spatial attention differ significantly. By introducing learnable weights, the model can adaptively enhance channel or spatial features, reducing information loss and preserving richer feature representations. After attention fusion, nonlinear transformations are applied to further enhance the feature representation ability. The enhanced attention mechanism can more effectively capture complex change features, particularly in high-resolution images, significantly improving the accuracy and robustness of change detection.

The structure of the DT-DEM is shown in Fig. 4. This module performs pixel-wise absolute difference operations on the input dual-time images $X_i$, where $i \in \{1, 2\}$, to preliminarily extract shallow change features $C_1$. In order to suppress background noise interference and obtain more accurate change features, the module incorporates the enhanced attention mechanism from Fig. 2. By dynamically adjusting the model's attention to different regions of the image, the mechanism enables a more focused enhancement of the truly changed areas, thereby generating the enhanced change features $E$

$$C_1 = |X_1 - X_2| \tag{1}$$

$$E = \text{Attention}(C_1). \tag{2}$$

In order to effectively fuse the enhanced change feature information back into the original diachronic feature map, the DT-DEM adopts a feature-weighted mapping strategy. Specifically, the enhanced feature $E$ is multiplied element-by-element with the input image $X_i$ to realize the weighted fusion of feature information. This process not only preserves the rich information of the original image, but also highlights the saliency of the changed region. To avoid the problem of gradient vanishing that may occur during deep network training, the DT-DEM module further introduces residual learning. The enhanced feature map is directly added to the original feature map $X_i$, and a convolutional module is applied for feature fusion and enhancement, resulting in the final representation $Z_i$, which contains rich long-range contextual information. This process not only facilitates the effective flow of information, but also significantly enhances the characterization of the object of change

$$Z_i = \text{Conv}(X_i \times E + X_i). \tag{3}$$

Finally, the more important change feature $C_2$ is extracted by performing another absolute difference operation on $Z_i$

$$C_2 = |Z_1 - Z_2|. \tag{4}$$

### D. Screening Functions Inverted Pyramid Network

The objects in remote sensing image change detection encompass a wide range of different target types, such as buildings, roads, and vegetation, which exhibit significant size differences. Effectively addressing the challenge of multiscale detection has become a core issue in change detection tasks and is crucial for improving the accuracy and comprehensiveness of change detection.

To achieve efficient learning and deep fusion of multiscale features, a S-FIPN was designed. Unlike traditional

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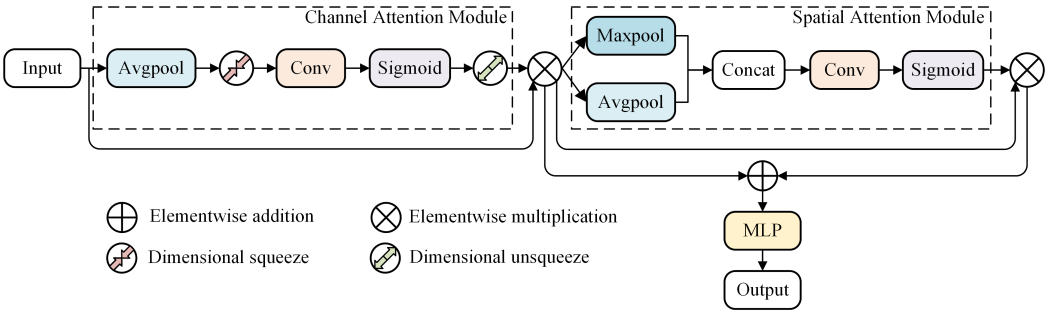



Fig. 3. Structure of the enhanced attention mechanism.

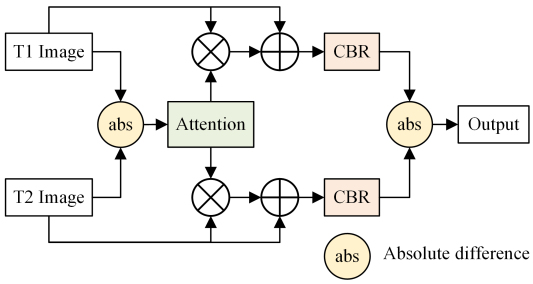



Fig. 4. Structure of the DT-DEM.

feature pyramids, which use bottom-up and top-down construction methods, S-FIPN adopts a top-down and bottom-up approach. S-FIPN utilizes the global semantic information from the top-level low-resolution features to guide the lower level features to focus on key change regions, enabling fine-grained extraction. This structure not only captures large-sized change targets quickly but also detects subtle changes with sensitivity, improving detection accuracy across all scales. Moreover, S-FIPN promotes the reverse flow and deep fusion of dual-temporal image information, enhancing local detail features and flexibly adapting to complex change scenarios, ensuring precise identification of every significant change. The model possesses a stronger capability to capture more comprehensive and accurate change detection features, significantly boosting the performance of change detection.

The network structure of S-FIPN is shown in Fig. 5. The S-FPN consists of two parts: feature fusion and feature selection. Feature fusion is used to merge multiscale features and obtain more comprehensive change feature information, while feature selection focuses on capturing more important change features to avoid interference from redundant information. In feature fusion, Input1 to Input5 are the five feature inputs containing different scales. These inputs are first processed through a series of CBR modules (CBR1 to CBR4) to adjust the channel dimensions, preparing them for subsequent feature fusion operations. Then, Basicblock1 to Basicblock3 are used to extract scale-specific feature information, deeply exploring and integrating features at different scales, generating richer feature representations. Feature selection is achieved through CBR5 to CBR8, which performs downsampling operations to gradually reduce the resolution of the feature maps while preserving key information. CBAM1 to CBAM5 are introduced as feature selectors to adaptively adjust the weight of the feature maps, allowing the network to focus more on change features, effectively improving detection accuracy. After feature selection, the feature maps undergo an upsampling operation to adjust the resolution to match that of the change detection head $(h/2, w/2)$, which then proceeds to the subsequent change detection task.

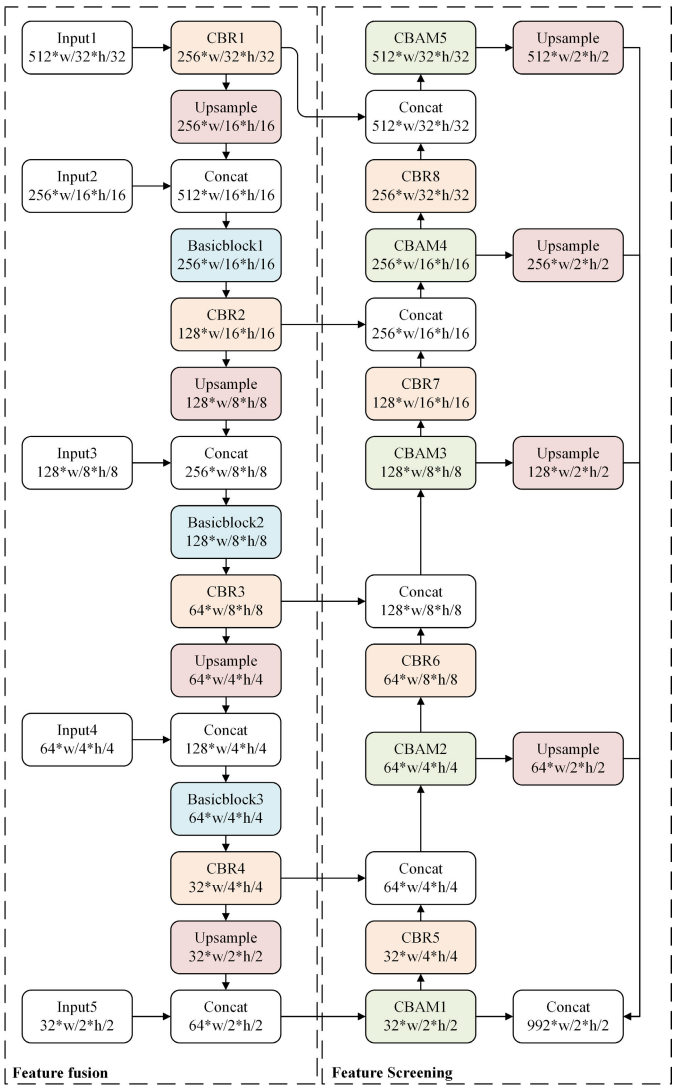



Fig. 5. Network architecture of S-FIPN, which contains two parts: feature fusion and feature screening.

## E. Loss Function

In the remote sensing image change detection task, focal loss [35] focuses on difficult samples and small targets detection by dynamically adjusting the sample weights, which

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TABLE I
DATASET STATISTICS AND SPLITS FOR LEVIR-CD, WHU-CD, NJDS, AND MSRS-CD

| Dataset | Resolution (m) | Train | Validation | Test |
|---|---|---|---|---|
| LEVIR-CD | 0.3 | 445 | 64 | 128 |
| WHU-CD | 0.2 | 336 | 48 | 96 |
| NJDS | 0.3 | 117 | 16 | 35 |
| MSRS-CD | 0.5 | 558 | 84 | 169 |

effectively mitigates the category imbalance problem due to the very low percentage of change regions, but its pixel-level optimization is prone to generating isolated noise. Dice coefficient loss [36], on the other hand, enforces connectivity and boundary smoothing over a wide range of change regions by maximizing the region overlap between the prediction and the true value, but may miss the detection of subtle changes. To address the problems of category imbalance and boundary optimization in the change detection task, this article combines focal loss and dice coefficient loss as a loss function. The complementary nature of the two makes the model perform better in detecting small targets, boundary regions, and global changes, while accelerating the convergence speed and improving the robustness of the model, which is especially suitable for scenarios with sparse change regions and complex distribution. Focal loss expression

$$L_{\text{fl}} = -(1 - p_t)^{\gamma} \log(p_t) \tag{5}$$

where $p_t$ is the prediction probability, when the label is a positive class, $p_t = p$, when the label is a negative class, $p_t = 1-p$. $1-p$ denotes the degree of difficulty in prediction, the smaller $p_t$ denotes the more difficult the sample is to classify. $Y$ is the moderating factor, which is used to control the weight of the easy-to-categorize sample. Dice coefficient loss expression

$$L_{\text{dice}} = 1 - \frac{2|X \bigcap Y|}{|X| + |Y|}. \tag{6}$$

In this particular example, $|X|$ represents the predicted outcome of the division, while $Y$ denotes the actual outcome. The symbol $|X \cap Y|$ indicates the intersection between the predicted result and the actual outcome, and $|X|+|Y|$ signifies the sum of the predicted result and the actual outcome. The combination of $L_f$ and $L_{\text{dice}}$ results in a loss of $L$

$$L = \alpha L_{\text{fl}} + \beta L_{\text{dice}}. \tag{7}$$

In order to reduce the complexity of hyper-parameter adjustment and accelerate the model convergence, a dynamic weight adjustment mechanism is introduced so that the weights are automatically adjusted according to the current loss, where

$$\alpha = \frac{L_{\text{fl}}}{L_{\text{fl}} + L_{\text{dice}}}, \quad \beta = \frac{L_{\text{dice}}}{L_{\text{fl}} + L_{\text{dice}}}$$

which helps the model to quickly find a reasonable optimization direction to improve the convergence speed, and the weights will be reduced when some kind of loss is dominant. When a loss is dominant, its weight will be reduced to avoid the model over-relying on that loss and ignoring other losses.

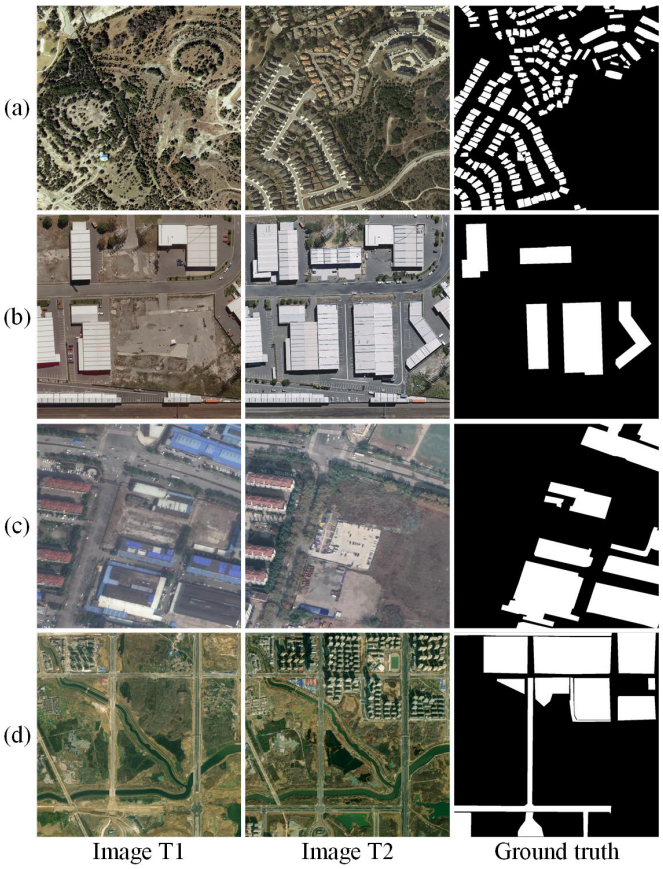



Fig. 6. Example plots of the four datasets LEVIR-CD, WHU-CD, NIDS, and MSRS-CD. Where (a) LEVIR-CD dataset, (b) WHU-CD dataset, (c) NJDS dataset, and (d) MSRS-CD dataset.

## IV. EXPERIMENTS

### A. Datasets

All experiments in this article are conducted on four datasets: LEVIR-CD [20], WHU-CD [37], NJDS [38], and MSRS-CD [39]. The detailed dataset division is shown in Table I, and specific examples are presented in Fig. 6. A brief introduction to these four datasets is provided below.

LEVIR-CD was proposed in 2020 and is primarily used for change detection in large-scale remote sensing images. The dataset includes 637 pairs of images, each with a size of 1024 × 1024 pixels. It encompasses a wide variety of building types, such as high-rise apartments, villa residences, and garages of various sizes. The images were captured from 20 different areas in several cities in Texas, USA, with a time span ranging from 2002 to 2018.

The WHU-CD dataset is a specialized dataset designed for building change detection. Its image size reaches 32 207 × 15 354, with images captured in Christchurch, New Zealand, in 2012 and 2016. Covering an area of 20.5 $km^2$, the dataset includes 12 796 buildings. Complementary to the LEVIR dataset, WHU-CD effectively validates the model's capability in building change detection from different perspectives, providing strong support for research.

NJDS was created by a research team at Nanjing Normal University to address height displacement issues in building change detection, targeting low-, mid-, and high-rise buildings.

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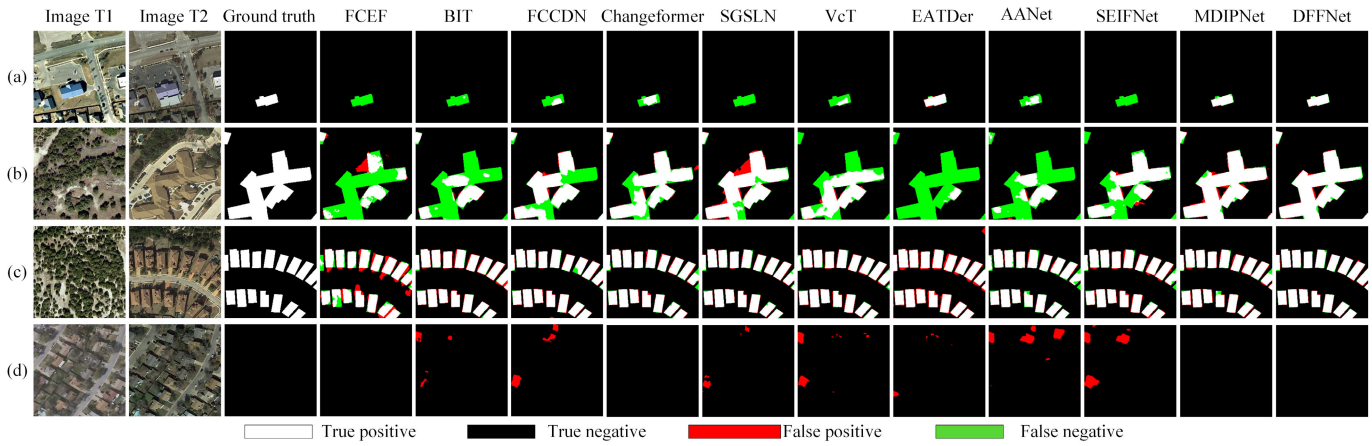



Fig. 7. Visual results of 11 algorithms tested on the LEVIR-CD dataset. Among them, (a)–(d) are the detection results on LEVIR-CD.

Its dual-temporal images were obtained from Nanjing City in 2014 and 2018, with an image size of 14 231 × 11 381 and a spatial resolution of 0.3 m. Notably, due to the relatively small amount of data involved, this dataset is well-suited to accurately evaluate the model's performance in low-data scenarios, providing critical insights for studies on model adaptability.

The MSRS-CD dataset consists of 841 pairs of remote sensing images, each with a size of 1024 × 1024 pixels and a spatial resolution of 0.5 m. The dataset includes various change targets such as newly constructed buildings, suburban expansion, vegetation changes, and road construction, showcasing the complexity of real-world change scenarios. The dataset is divided into training, validation, and test sets in a 7:1:2 ratio.

### B. Experimental Environment and Evaluation Indicators

The model training was implemented on the PyTorch 1.13.1 framework using an NVIDIA Tesla L40 GPU with 48 GB of memory. The optimizer used during training was AdamW, with a weight decay parameter set to 0.001. The training process spanned 300 epochs, with an initial learning rate of 0.0002 and a batch size of 8. Validation was initiated after 150 epochs of training. If the validation set showed no improvement for 12 consecutive epochs, the learning rate was reduced by a factor of 10. If no improvement was observed after 50 epochs, training was terminated. This training configuration efficiently utilized GPU resources to achieve optimal training performance and resource utilization.

The evaluation metrics selected for the change detection algorithms in this article are precision ($P$), recall ($R$), intersection over union (IoU), overall accuracy (OA), and $F1$ score. Among these, precision focuses on the accuracy of positive predictions made by the model, while Recall emphasizes the model's ability to capture positive instances. IoU is a comprehensive metric that measures the overlap between predicted and actual regions. OA evaluates the overall classification accuracy of the model, and the $F1$ score balances the tradeoff between precision and recall. The calculations for these evaluation metrics are as follows:

$$\text{Precision} = \frac{\text{TP}}{(\text{TP}+\text{FP})} \tag{8}$$

$$\text{Recall} = \frac{\text{TP}}{\text{FN}+\text{TP}} \tag{9}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP}+\text{FN}+\text{FP}} \tag{10}$$

$$\text{OA} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FN}+\text{FP}} \tag{11}$$

$$F1 = \frac{2}{\text{Recall}^{-1}+\text{Precision}^{-1}}. \tag{12}$$

Here, TP represents the number of samples predicted as changed and actually changed, TN denotes the number of samples predicted as unchanged and actually unchanged, FP refers to the number of samples predicted as changed but actually unchanged, and FN indicates the number of samples predicted as unchanged but actually changed.

### C. Comparative Experiments With SOTA Algorithms

In order to comprehensively and accurately evaluate the comprehensive performance of DFFNet, this article compares it with eight cutting-edge algorithms, namely, FCEF, BIT, FCCDN, Changeformer, SGSLN, VcT, EATDer, and AANet on four datasets, namely, LEVIR, NJDS, WHU, and MSRS. The eight cutting-edge algorithms are briefly described below.

1) *FCEF [16]:* As a classic work in the field of change detection, FCEF constructs a fully convolutional change detection network architecture. The network adopts a UNet structure to extract change features from bi-temporal images.
2) *BIT [21]:* This network effectively utilizes the local feature extraction capability of convolution and the global feature extraction capability of the Transformer. It designs a change detection network that combines local precision with a global perspective, addressing complex and variable detection scenarios.
3) *FCCDN [40]:* This network is composed of dual encoder–decoder structures and employs a self-supervised learning strategy to constrain feature

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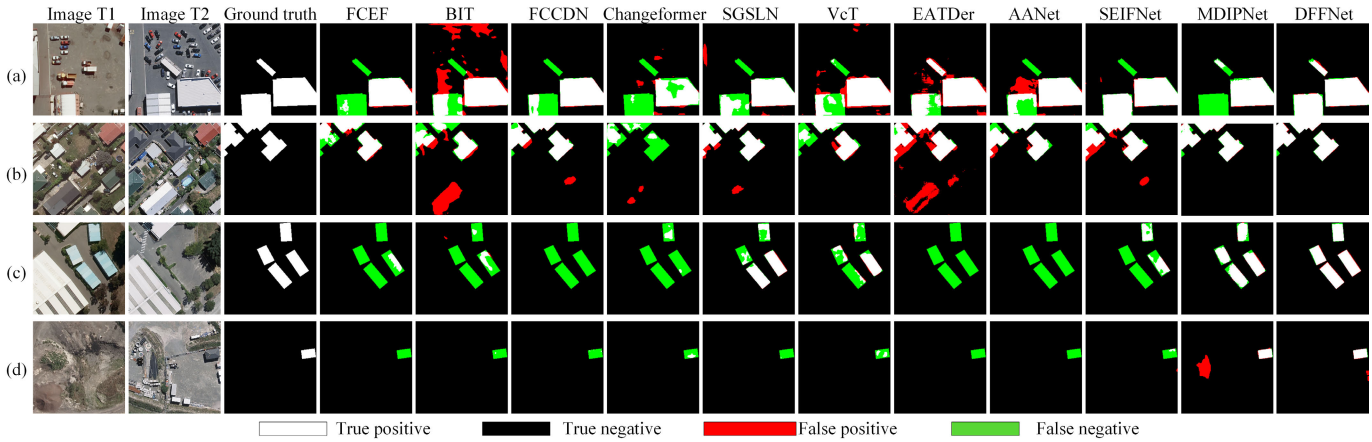



Fig. 8. Visual results of 11 algorithms tested on the WHU-CD dataset. Among them, (a)–(d) are the detection results on WHU-CD.

learning, resulting in better extraction of change feature information.

4) *Changeformer [24]:* As a bold attempt to apply a fully Transformer-based architecture in change detection, Changeformer abandons traditional structural components. It fully leverages the powerful performance of Transformers to perform comprehensive feature extraction and analysis of bi-temporal images, showcasing unique advantages in handling complex change relationships.
5) *SGSLN [41]:* This network innovatively integrates channel exchange techniques into the change detection process. Through a meticulously designed channel exchange mechanism, it breaks conventional limitations during feature extraction and processing, uncovering more hidden change information within the data and providing a new approach for precise detection.
6) *VcT [22]:* This network combines graph neural networks to extract change feature information from bi-temporal images. By integrating graph neural networks, VcT efficiently extracts correlated and deeper change feature information, enhancing detection accuracy and reliability.
7) *EATDer [42]:* This network introduces an innovative edge decoder to effectively extract edge information of change features. This design enables the network to focus on the edge information of change features during processing. By precisely extracting these edge details, it refines the overall change detection results and improves their interpretability.
8) *AANet [43]:* The core of AANet lies in the full utilization of multiscale information. Through clever design, it mines and integrates multiscale information contained in different layers, enabling comprehensive capture of change monitoring information.
9) *SEIFNet [44]:* SEIFNet improves the network's ability to reproduce features of changing objects through spatio-temporal enhancement and interlayer fusion techniques.
10) *MDIPNet [45]:* MDIPNet simplifies multilayer multihead attention into single-layer single-head structure, designs lightweight parallel fusion module to achieve efficient feature integration, and builds cross-space perception module to enhance semantic consistency.

On the LEVIR-CD dataset, the comparative experimental results with current SOTA algorithms are presented in Table II. The data clearly demonstrate that DFFNet achieves the best performance in four key evaluation metrics: $P$, $F1$, IoU, and OA, with specific values of 92.97%, 92.50%, 85.74%, and 99.24%, respectively. This outstanding performance is attributed to DFFNet's superior capability in feature extraction and fusion. It is worth noting that although DFFNet's $R$ is slightly lower than that of the second-ranked EATDer by 0.93%, this is because EATDer incorporates an additional boundary detection head to improve recall. However, this also introduces more false positives, resulting in a Precision that is 7.24% lower than DFFNet. Fig. 7 visually compares the inference results of the nine algorithms on the LEVIR-CD dataset. Fig. 7(a) and (b) shows that DFFNet has less miss detection for both large-size and small-size targets, which is attributed to the powerful full-size feature enhancement and fusion capability of DFFNet's S-FIPN module. Fig. 7(c) vividly highlights the significant advantages of DFFNet in boundary detection. Fig. 7(d) further demonstrates its excellent performance in reducing false alarms, which is attributed to the powerful change feature resolution capability of DT-DEM.

On the WHU-CD dataset, the comparative results are shown in Table III. The experimental data again validate the superior performance of DFFNet, which achieves the best results across all four evaluation metrics: $R$, $F1$, IoU, and OA, with values of 85.63%, 90.55%, 82.65%, and 99.14%, respectively. Fig. 8(a) further demonstrates the superiority of DFFNet for edge detection. The visualization results in Fig. 8(b)–(d) further support these findings, illustrating that DFFNet exhibits clear advantages in reducing both false positives and false negatives compared to other algorithms.

On the NJDS dataset, the comparative results are displayed in Table IV. DFFNet achieves the best performance in four key metrics: $R$, $F1$, IoU, and OA, with respective values of 76.91%, 81.04%, 68.02%, and 97.64%. The NJDS dataset has a small amount of data, and the visualization results in Fig. 9 further show that, in time, with relatively limited data, DFFNet

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TABLE II
COMPARISON WITH OTHER SOTA MODELS ON THE LEVIR-CD DATASET, WHERE THE BOLDED VALUES ARE THE BEST

| Methods | P (%) | R (%) | F1 (%) | IoU (%) | OA (%) |
|---|---|---|---|---|---|
| FCEF | 87.73 | 82.02 | 84.78 | 66.36 | 98.50 |
| BIT | 90.33 | 89.56 | 89.94 | 81.72 | 98.98 |
| FCCDN | 92.17 | 90.49 | 91.32 | 83.04 | 99.12 |
| Changeformer | 92.05 | 88.80 | 90.40 | 82.48 | 99.04 |
| SGSLN | 92.63 | 90.71 | 91.66 | 83.48 | 99.16 |
| VcT | 92.57 | 87.65 | 89.93 | 81.51 | 98.99 |
| EATDter | 85.73 | **92.97** | 89.20 | 80.51 | 98.85 |
| AANet | 92.10 | 87.74 | 89.87 | 81.60 | 98.99 |
| SEIFNet | 91.82 | 88.03 | 89.89 | 81.63 | 98.99 |
| MDIPNet | 92.04 | 90.22 | 91.12 | 83.69 | 99.14 |
| DFFNet | **92.97** | 92.04 | **92.50** | **85.74** | **99.24** |

TABLE III
COMPARISON WITH OTHER SOTA MODELS ON THE WHU-CD DATASET, WHERE THE BOLDED VALUES ARE THE BEST

| Methods | P (%) | R (%) | F1 (%) | IoU (%) | OA (%) |
|---|---|---|---|---|---|
| FCEF | 92.68 | 80.74 | 86.30 | 68.54 | 98.76 |
| BIT | 78.48 | 84.13 | 81.21 | 68.46 | 98.12 |
| FCCDN | 95.24 | 83.58 | 89.03 | 79.85 | 99.01 |
| Changeformer | 88.08 | 76.92 | 82.12 | 69.67 | 98.38 |
| SGSLNe | 95.53 | 84.38 | 89.61 | 80.79 | 99.06 |
| VcT | 83.07 | 80.09 | 81.55 | 68.85 | 98.25 |
| EATDer | 87.97 | 85.48 | 86.71 | 76.53 | 98.73 |
| AANet | 77.12 | 80.97 | 79.00 | 65.29 | 97.92 |
| SEIFNet | 88.72 | 82.08 | 85.27 | 74.33 | 98.63 |
| MDIPNet | **96.44** | 82.76 | 89.08 | 80.31 | 99.02 |
| DFFNet | 96.06 | **85.63** | **90.55** | **82.65** | **99.14** |

TABLE IV
COMPARISON WITH OTHER SOTA MODELS ON THE NJDS DATASET, WHERE THE BOLDED VALUES ARE THE BEST

| Methods | P (%) | R (%) | F1 (%) | IoU (%) | OA (%) |
|---|---|---|---|---|---|
| FCEF | 69.97 | 38.10 | 49.34 | 29.52 | 94.86 |
| BIT | 71.81 | 55.79 | 62.79 | 45.77 | 95.66 |
| FCCDN | **87.79** | 48.73 | 62.68 | 45.38 | 96.19 |
| Changeformer | 85.53 | 31.59 | 46.14 | 29.99 | 95.15 |
| SGSLN | 83.59 | 58.42 | 68.77 | 52.22 | 96.51 |
| VcT | 80.14 | 46.15 | 58.57 | 41.41 | 95.71 |
| EATDer | 61.63 | 50.85 | 55.61 | 38.52 | 94.67 |
| AANet | 79.57 | 54.19 | 64.47 | 47.57 | 96.08 |
| SEIFNet | 85.66 | 54.27 | 66.45 | 49.75 | 96.40 |
| MDIPNet | 85.90 | 59.98 | 70.64 | 54.61 | 96.72 |
| DFFNet | 85.64 | **76.91** | **81.04** | **68.02** | **97.64** |

TABLE V
COMPARISON WITH OTHER SOTA MODELS ON THE MSRS-CD DATASET, WHERE THE BOLDED VALUES ARE THE BEST

| Methods | P (%) | R (%) | F1 (%) | IoU (%) | OA (%) |
|---|---|---|---|---|---|
| FCEF | 74.99 | 66.69 | 70.59 | 45.44 | 91.80 |
| BIT | 75.73 | 70.79 | 73.18 | 57.70 | 92.34 |
| FCCDN | 75.56 | 71.31 | 73.37 | 55.42 | 92.36 |
| Changeformer | 72.22 | 72.94 | 72.58 | 56.96 | 91.86 |
| SGSLNe | 77.39 | 69.73 | 73.36 | 56.28 | 92.52 |
| VcT | 76.64 | 69.72 | 73.02 | 57.50 | 92.39 |
| EATDer | 66.73 | **84.17** | 74.44 | 59.29 | 91.47 |
| AANet | 71.94 | 77.03 | 74.40 | 59.23 | 92.17 |
| SEIFNet | 75.24 | 75.46 | 75.24 | 60.30 | 92.67 |
| MDIPNet | **81.14** | 70.24 | 75.30 | 60.38 | 93.19 |
| DFFNet | 76.22 | 78.69 | **77.44** | **61.82** | **93.23** |

still performs robustly and still provides reliable detection in terms of edge detection effectiveness and missed detection errors.

Finally, on the MSRS-CD dataset, the quantitative analysis results of the SOTA comparison are shown in Table V. The experimental data indicate that DFFNet achieves the highest performance in three critical metrics: $F1$ score, IoU, and OA, with values of 77.44%, 61.82%, and 93.23%, respectively. The visual comparisons in Fig. 10 further highlight DFFNet's strengths in edge detection and its remarkable ability to reduce false negatives. EATDer achieved the highest Recall due to the fact that EATDer additionally integrates an edge detection head that is able to obtain more edge information. This edge detection head captures most of the true positive class pixels (high recall), but also incorrectly predicts many negative class pixels as positive (high FP, resulting in low precision). This can also be derived from Fig. 10(c).

In summary, whether under scenarios with abundant data or relatively limited datasets, DFFNet consistently demonstrates exceptional change detection capabilities. Its performance is particularly noteworthy in edge information extraction and in reducing both false positives and false negatives, firmly establishing its broad applicability and superior effectiveness across various change detection tasks.

### D. Ablation Studies

Ablation experiments were conducted on the LEVIR-CD and MSRS-CD datasets to verify the efficient performance of DT-DEM and AS-HPN. Given that dimension addition is a commonly adopted method for difference calculation, it was chosen as an alternative to DT-DEM for comparison. At the same time, the feature pyramid network (FPN), a classic method for feature fusion, was used to replace AS-HPN to evaluate its effectiveness. The results in Table VI show that when the DT-DEM module is introduced alone, the $F1$ score increased by 0.82% and the IoU by 1.49% on the LEVIR-CD dataset. On the MSRS-CD dataset, the $F1$ score and IoU increased by 0.57% and 1.42%, respectively. Notably, by comparing Fig. 11(e) with Fig. 11(d), a significant improvement in the omission of change targets can be clearly observed, which strongly proves the exceptional ability of DT-DEM in capturing change detection features.

Furthermore, when only the AS-HPN module is added, the $F1$ score and IoU on the LEVIR-CD dataset achieved a substantial increase of 3.06% and 5.29%, respectively. On the MSRS-CD dataset, these two metrics also improved by 1.82% and 2.27%. By comparing Fig. 11(f) with Fig. 11(d), it is intuitively clear that AS-HPN performs better in edge detection when handling targets of different sizes, with a noticeable reduction in both false detection and missed detection. This strongly validates the powerful capability of AS-HPN in full-size feature extraction and fusion, particularly in detecting small changes and weak edge information.

When both DT-DEM and AS-HPN modules are integrated, the experimental results reached their optimal performance. On the LEVIR-CD dataset, the $F1$ score and IoU achieved excellent results of 92.50% and 85.73%, respectively. This

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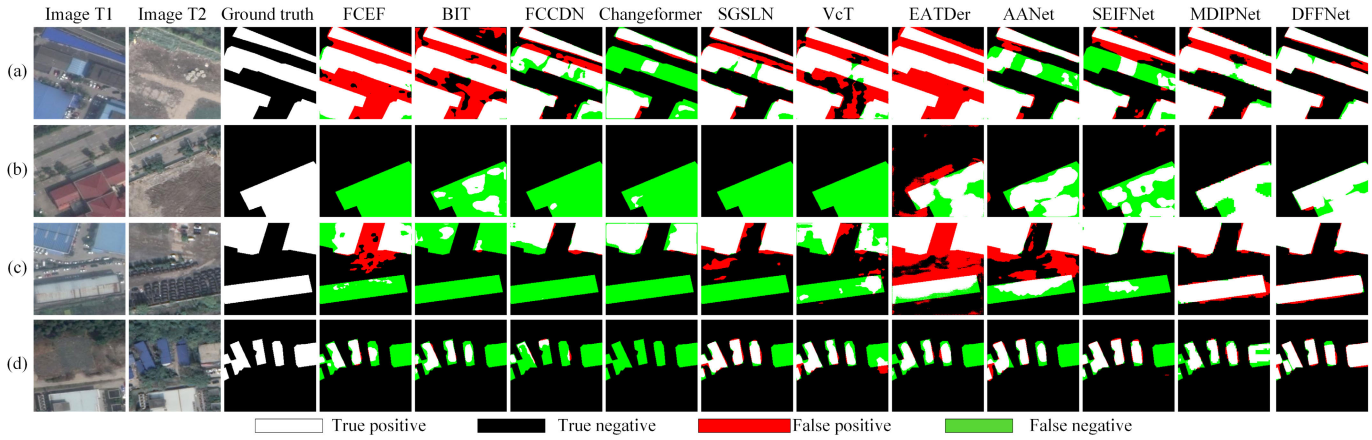



Fig. 9. Visual results of 11 algorithms tested on the NJDS dataset. Among them, (a)–(d) are the detection results on NJDS.

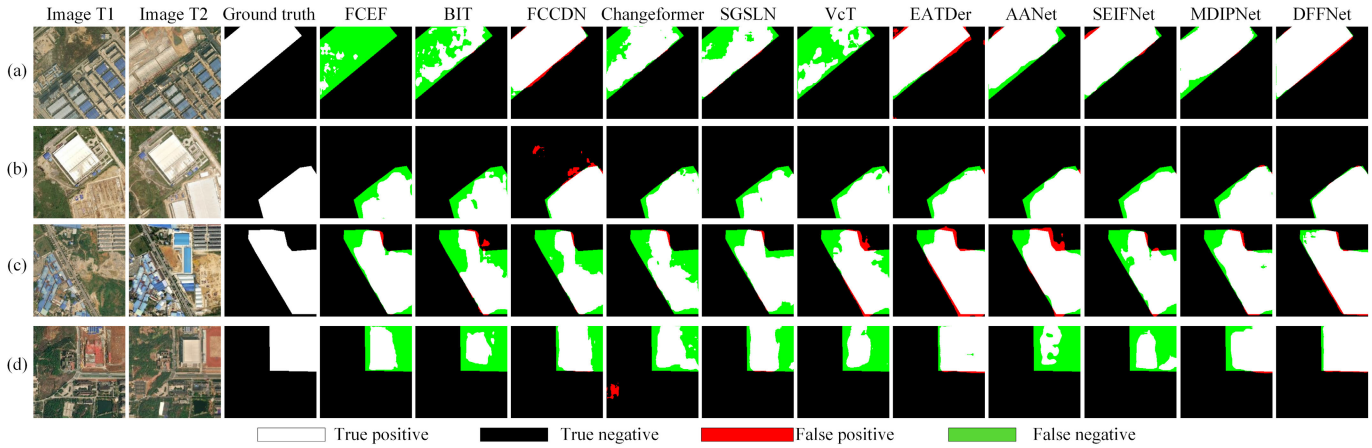



Fig. 10. Visual results of 11 algorithms tested on the MSRS-CD dataset. Among them, (a)–(d) are the detection results on MSRS-CD.

TABLE VI

RESULTS OF ABLATION EXPERIMENTS ON THE LEVIR-CD DATASET AND THE MSRS-CD DATASET

| Add | DT-DEM | FPN | S-HPN | LEVIR-CD | | MSRS-CD | |
|---|---|---|---|---|---|---|---|
| | | | | F1 | IoU | F1 | IoU |
| ✓ | ✗ | ✓ | ✗ | 88.69 | 79.14 | 74.57 | 57.88 |
| ✗ | ✓ | ✓ | ✗ | 89.51 | 80.63 | 75.14 | 59.30 |
| ✓ | ✗ | ✗ | ✓ | 91.75 | 84.43 | 76.39 | 60.15 |
| ✗ | ✓ | ✗ | ✓ | 92.50 | 85.74 | 77.44 | 61.82 |

clearly demonstrates that the synergistic effect between the two modules can significantly enhance the overall performance of the model in change detection tasks. Similarly, on the MSRS-CD dataset, the addition of both modules led to further improvements in the $F1$ score and IoU. Combined with the outstanding performance in Fig. 11(g), it can be verified that the integration of DT-DEM and AS-HPN maximizes the accuracy and segmentation effectiveness of the model in change detection tasks.

### *E. Analysis of Model Complexity and Inference Time*

As shown in Table VII, the computational complexity and reasoning efficiency of DFFNet are analyzed in comparison with current mainstream methods based on the MSRS-CD dataset. In terms of computational complexity, DFFNet's FLOPs (56.93 G) are higher than lightweight architectures such as BIT (8.75 G) and FCCDN (12.49 G), but lower than the computationally intensive model Changeformer

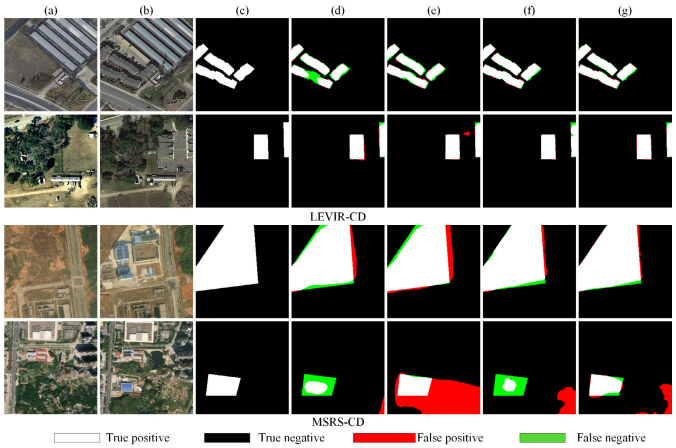



Fig. 11. Visual results of the fusion experiment. (a) Image t1. (b) Image t2. (c) Ground truth. (d) Add + FPN. (e) DT-DEM + FPN. (f) Add + S-HPN. (g) DT-DEM + S-HPN.

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TABLE VII
RESULTS OF COMPLEXITY AND INFERENCE TIME ANALYSIS OF DIFFERENT NETWORKS ON MSRS-CD DATASET

| Methods | F1(%) | FLOPs(G) | Param(M) | Inference time(ms) |
|---|---|---|---|---|
| FCEF | 70.59 | 3.55 | 1.35 | 1.90 |
| BIT | 73.18 | 8.75 | 3.04 | 11.58 |
| FCCDN | 73.37 | 12.49 | 6.31 | 15.29 |
| Changeformer | 72.58 | 202.79 | 41.03 | 28.78 |
| SGSLN | 73.36 | 11.50 | 6.04 | 10.31 |
| VcT | 73.02 | 10.64 | 3.57 | 14.68 |
| EATDer | 74.44 | 23.46 | 6.61 | 21.18 |
| AANet | 74.40 | 24.21 | 15.82 | 10.63 |
| SEIFNet | 75.24 | 8.37 | 27.91 | 11.54 |
| MDIPNet | 75.30 | 14.10 | 40.21 | 23.58 |
| DFFNet | 77.44 | 56.93 | 23.32 | 19.63 |

(202.79 G). This phenomenon stems from the fact that the multistage feature fusion mechanism designed by DFFNet needs to perform cross-layer feature alignment and enhancement operations, resulting in an elevated computational density. In the parameter count dimension, its model size (23.32 M) is in the middle of the range, significantly lower than models with large parameter counts such as MDIPNet (40.21 M), but higher than lightweight networks such as BIT (3.04 M), indicating that the network avoids parameter redundancy while maintaining structural complexity. Notably, despite the high computational complexity, DFFNet exhibits excellent inference efficiency (19.63 ms), which is 31.7% faster than Changeformer (28.78 ms) and 16.8% faster than MDIPNet (23.58 ms). Combined with the detection performance, DFFNet maintains a significant advantage in the $F1$ metric (77.44%), proving that it achieves an effective balance between high computational complexity and detection accuracy. This feature makes it suitable for application scenarios with stringent requirements on detection accuracy and moderate computational resources.

## V. DISCUSSION

### A. Loss Function

To verify the effectiveness of joint losses, we conducted the comparison experiments in Table VIII, which show that the lowest $F1$ values are achieved when only one loss is involved in training. Compared with fixed weights, dynamic weight adjustment achieved the highest $F1$ value. This is due to the significant advantage of dynamic weight adjustment in adaptively balancing multiple loss targets. It automatically adjusts the loss weights according to the current magnitude of the loss weights, prioritizing major losses at an early stage to accelerate convergence, while suppressing over-reliance on any single objective. This mechanism ensures gradient balancing, prevents local optima, and reduces the complexity of hyperparameter tuning by eliminating manual weight searches.

Although the current work uses classical loss combinations, the emerging SNIB [46] can enhance the robustness of the model to noise interference through nonlinear information bottleneck constraints, while SMEIL [47] and MeMEE [48] show excellent meta-learning capabilities in impulse neural networks, and SeLHIB [49] can provide robust estimation of optical flow in the presence of noise. These methods provide new ideas to improve the domain adaptation and sparse sample learning of change detection models, which will be the focus of subsequent research.

TABLE VIII
COMPARATIVE EXPERIMENTAL RESULTS FOR DIFFERENT WEIGHT VALUES

| $L_{fl}$ | $L_{dice}$ | F1(%) |
|---|---|---|
| 0 | 1.0 | 91.89 |
| 0.3 | 0.7 | 92.04 |
| 0.5 | 0.5 | 92.12 |
| 0.7 | 0.3 | 91.75 |
| 1.0 | 0 | 91.23 |
| $\frac{L_{fl}}{L_{fl}+L_{dice}}$ | $\frac{L_{dice}}{L_{fl}+L_{dice}}$ | 92.50 |

### B. Model Generalization

Despite the current experimental focus on optical imagery, the mechanism for modeling temporal phase differences in DT-DEM has theoretical similarities with coherence analysis for SAR change detection. Initial tests on the Ottawa SAR dataset showed a KC of 93.5% (uncorrelated) for DFFNet, validating the potential cross-domain adaptability of the architecture. Complete cross-modal experiments will be advanced as a focus of future work.

## VI. CONCLUSION

This article proposes a deep feature fusion-based high-resolution remote sensing image full-size change detection network (DFFNet), which demonstrates outstanding performance in full-size feature extraction and edge information enhancement. It effectively addresses the limitations of existing methods in processing small-sized targets and weak-edge feature information. The designed DT-DEM can comprehensively extract multiscale change information, significantly improving the ability to capture subtle changes and edge features in complex scenarios. The S-FIPN further enhances the detection of fine-grained changes and weak edge information by efficiently extracting and fusing full-size features.

Experiments on four publicly available datasets, including LEVIR-CD, WHU-CD, NJDS, and MSRS-CD, show that DFFNet outperforms ten advanced algorithms in terms of $F1$ score and IoU across all datasets. These results fully validate the effectiveness and superiority of DFFNet in high-resolution remote sensing image change detection tasks, providing strong support for further research and practical applications.

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